

Rayyan Ahmed

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PROFESSIONAL SUMMARY

Generative AI Engineer with 2+ years of experience turning operational problems into end-to-end agentic AI solutions. Translates business requirements into working systems by designing retrieval and data flows, orchestrating agents and tools, integrating cloud and local models, and implementing APIs, evaluation, guardrails, and human review. Builds reliable, auditable workflows across healthcare, mental health, revenue cycle management (RCM), medical coding, financial analytics, document intelligence, and browser automation using Python, FastAPI, LangGraph, MCP, Transformers, AWS Bedrock, and vLLM.

SKILLS

Generative AI & Agentic Systems: Harness Engineering, LLM applications, AI agents, multi-agent orchestration, LangGraph, LangChain, MCP, tool/function calling, structured outputs, Deep Agents, MCP Servers and context engineering

RAG, Evaluation & Safety: Retrieval-augmented generation, hybrid retrieval (BM25 + FAISS), embeddings, semantic search, grounding, LLM evaluation, guardrails, validation, provenance, human-in-the-loop

Models, Fine-Tuning & Inference: PyTorch, Transformers, Hugging Face, VLMs/multimodal AI, LoRA/QLoRA, PEFT, quantization, llama.cpp, vLLM, LiteLLM, AWS Bedrock

LLMOps & Deployment: Python, SQL, TypeScript, FastAPI, model APIs, Redis, Celery, SSE/WebSockets, PostgreSQL, Docker, Kubernetes, CI/CD, AI Observability, Lang Fuse and Post Hog.

Automation & Domains: Computer-use agents, Playwright, Browser Use, RCM, healthcare finance, medical coding, document intelligence, financial analytics

Core Strengths: Communication, ownership, execution, collaboration, problem solving, prioritization, adaptability

PROJECTS

- **Prism** (Private) **Agentic RAG • MCP • SQL-Grounded LLMs • Evaluation & Guardrails**
Built an agentic GenAI platform for **Finance** using Text to SQL-grounded RAG and tool calling, reducing recurring team analysis from hours to minutes and producing decision-ready financial insights and reports.
Separated deterministic calculations from LLM reasoning and added capability-aware routing, structured outputs, provenance, evaluation gates, and human review to keep financial analysis auditable and reduce hallucination risk.
- **MedCodex** (Private) **LangGraph • Multi-Agent Systems • AI Observability • LLM OPS**
Built a LangGraph multi-agent workflow that extracts clinical context, proposes ICD-10-CM, CPT, and HCPCS candidates, validates them against available references, and applies consistency checks with retry and escalation paths.
Improved medical-coding workflow efficiency by approximately 40%, shifting the team from manual drafting to reviewing and refining AI-generated coding outputs; implemented provider-agnostic routing, PHI masking, structured outputs, and audit trail compliance.
- **Auto Apply AI** **Computer-Use Agents • Browser Automation • RAG • Multi-Model Routing**
Built a browser-use agent platform for job discovery, semantic resume-job matching, ATS scoring, tailored resume and cover-letter generation, and human-approved application workflows.
Orchestrated Browser Use and Playwright agents with LiteLLM and Bedrock routing, FAISS retrieval, Redis queues, WebSocket streaming, and CAPTCHA/2FA handoff for observable, resilient browser automation.

EXPERIENCE

- AI Engineer: Nobility RCM | Islamabad, Pakistan** **Mar 2025 – Present**
- Design and deliver Generative AI systems for healthcare and RCM, including agentic financial analytics, medical coding, grounded question answering, document intelligence, and browser automation.
 - Architect LangGraph and MCP workflows with agent state, tool calling, structured outputs, hybrid retrieval, retry logic, evaluation gates, and human review; connect them to governed billing and financial data through FastAPI.
 - Engineer cloud and local inference paths with prompt and context engineering, provider-aware routing, PHI-aware controls, provenance, and deterministic calculations for reliable, auditable financial outputs.
- Data Analyst: OZIBOOK Tech Solutions | Remote, India** **Oct 2024 – Jan 2025**
- Built preprocessing, schema-validation, class-balancing, and feature-engineering pipelines for claim-anomaly modeling; benchmarked Random Forest, XGBoost, and Logistic Regression to support model selection.
- Data Scientist (Machine Learning): Digital Empowerment Networks | Islamabad, Pakistan** **Jul – Sep 2024**
- Developed and tuned business-KPI forecasting models with Pandas, NumPy, and scikit-learn, improving repeatability through structured feature pipelines, model evaluation, and hyperparameter tuning.

ACHIEVEMENTS

Third Prize - Qwen-Image LoRA Training Competition: 1.7 GB LoRA, 12,000+ images, 40 categories, single-GPU training

EDUCATION

FAST NUCES Islamabad | Islamabad, Pakistan **Aug 2022 – Jun 2026**
BS Software Engineering